

SPENCER, IA, USA

WMO: 726500

Lat: 43.164N

Lon: 95.202W

Elev: 1338

StdP: 14.00

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 14972

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.4	-6.7	-17.9	2.2	-10.7	-13.1	2.8	-6.0	34.1	20.0	30.8	18.7	7.9	280	0.645

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.6	91.2	74.4	88.8	73.1	86.0	71.5	77.9	86.9	75.9	85.0	74.2	82.7	14.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.1	138.4	83.6	73.0	128.6	81.6	71.6	122.5	80.1	42.3	86.9	40.4	84.8	38.6	82.3	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
29.5	26.3	23.7	-20.9	96.4	5.1	2.7	-24.6	98.4	-27.5	100.0	-30.4	101.5	-34.1	103.5		
			-21.3	81.2	4.9	1.9	-24.9	82.5	-27.7	83.6	-30.5	84.7	-34.1	86.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.8	17.0	20.6	33.6	47.1	59.0	69.3	73.2	69.9	63.4	49.3	35.2	21.8
	DBStd	22.14	12.89	12.80	12.55	10.24	8.85	6.86	6.02	5.72	8.60	9.72	10.78	12.07
	HDD50	4032	1024	824	525	170	22	0	0	0	7	131	454	875
	HDD65	7398	1489	1244	973	542	230	35	8	19	131	494	893	1340
	CDD50	2845	0	0	17	82	300	579	719	618	408	110	12	0
	CDD65	735	0	0	0	4	43	163	262	172	83	8	0	0
	CDH74	7578	0	0	7	101	540	1683	2661	1578	880	126	2	0
	CDH80	2620	0	0	1	29	176	598	1015	485	283	33	0	0
	WSAvg	11.1	11.8	12.2	12.5	13.7	12.5	10.4	8.3	7.9	9.4	11.0	11.7	11.6
	PrecAvg	29.3	0.6	0.6	1.6	3.1	4.1	4.6	3.7	3.8	2.9	2.2	1.3	0.8
(15) Precipitation	PrecMax	43.0	1.2	2.0	3.4	6.5	6.5	10.4	6.9	9.7	12.9	6.3	4.0	1.9
	PrecMin	21.7	0.0	0.1	0.0	0.9	1.1	1.6	0.6	1.2	0.2	0.5	0.0	0.1
	PrecStd	5.1	0.3	0.4	0.9	1.5	1.5	2.4	1.7	2.1	2.6	1.5	1.1	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.0	55.3	73.2	84.5	91.0	94.2	95.0	92.6	90.8	84.7	70.5	54.2
		MCWB	41.0	48.6	57.2	62.8	68.5	73.6	77.2	76.0	71.4	65.1	55.4	48.9
	2%	DB	41.0	46.8	64.9	75.9	84.6	90.0	91.3	88.5	87.5	77.0	63.2	46.5
		MCWB	36.5	42.3	54.0	58.2	65.2	71.9	76.6	74.7	70.7	61.5	52.8	40.8
	5%	DB	36.6	41.1	57.5	70.5	80.4	86.5	88.9	84.9	82.9	71.4	56.9	40.7
		MCWB	33.7	37.4	50.0	55.3	63.2	70.2	75.3	72.7	67.8	59.3	48.4	36.6
	10%	DB	33.8	36.8	51.8	64.8	75.2	83.0	85.8	82.1	79.3	65.9	51.8	36.9
		MCWB	31.7	34.1	46.1	52.6	60.7	68.7	73.2	70.7	66.0	55.8	45.4	34.1
	0.4%	WB	41.6	49.8	60.8	64.4	72.0	77.5	81.6	79.4	75.1	69.6	58.7	51.0
		MCDB	48.6	56.4	67.5	78.6	84.1	88.2	89.7	88.5	85.9	78.6	67.2	54.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.0	42.2	55.5	60.4	68.3	75.0	78.7	76.8	72.9	64.5	53.7	41.3
		MCDB	40.5	46.7	63.5	72.8	80.3	85.7	87.4	85.5	82.6	71.8	60.9	46.7
	5%	WB	34.0	37.7	51.0	57.4	65.6	72.8	77.0	74.6	70.8	60.9	49.8	37.2
		MCDB	36.3	41.0	57.4	67.6	76.2	82.7	85.9	82.4	79.4	69.0	56.1	40.5
	10%	WB	31.7	34.3	45.6	53.9	63.0	70.8	75.1	72.7	68.5	57.0	45.3	34.3
		MCDB	33.6	36.6	51.9	62.3	72.0	80.4	83.8	79.8	76.3	65.3	51.5	36.7
(35) Mean Daily Temperature Range	5% DB	MDBR	17.5	17.6	18.4	22.6	22.3	21.5	21.6	21.8	24.8	23.0	20.3	16.9
		MCDBR	21.2	21.2	29.0	32.4	29.4	25.3	22.9	23.8	27.4	30.3	29.4	22.3
		MCWBR	17.4	16.7	18.7	17.4	13.7	11.6	11.2	12.4	13.1	16.6	19.1	17.4
	5% WB	MCDBR	19.3	19.9	27.5	27.5	24.5	21.3	20.9	21.1	22.8	25.1	26.4	21.0
		MCWBR	16.0	15.9	18.2	16.2	13.0	11.7	11.7	12.3	12.2	15.7	18.3	16.8
(40) Clear-Sky Solar Irradiance	taub		0.271	0.287	0.314	0.363	0.389	0.405	0.416	0.409	0.367	0.330	0.290	0.273
	taud		2.433	2.405	2.428	2.321	2.318	2.332	2.284	2.288	2.375	2.467	2.503	2.459
	Ebn at Noon		278	291	295	286	280	275	271	269	273	271	269	266
	Edh at Noon		23	29	32	39	40	39	41	40	34	27	22	21
(44) All-Sky Solar Radiation	RadAvg		598	939	1216	1510	1765	1960	2081	1733	1397	943	640	476
	RadStd		59	85	110	135	126	146	94	123	92	119	44	43

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (53 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page